

Carlo M. Vilches

Research Economist — Energy Markets, Regulation & Quantitative Modeling

Barcelona, Spain | carlomvilc@gmail.com | +34 687 760 513 | github.com/cmldata | cmldata.github.io/cv

Spanish nationality · mobile

PROFILE

Economist (10+ years) specialised in sectoral regulation, competition and tariff design, used to defending strategic interests before regulatory authorities. I build economic models for forecasting and impact assessment, cost the compensation of public service obligations and subsidy schemes, and communicate results clearly to senior management and institutional decision-makers. Proficient in statistical and econometric tools (Python, R) and in Excel. Fluent French (C1) and English (C1).

SKILLS & METHODS

Languages (prog.): Python · R · SQL · SAS · Stata

Office / BI: Excel (advanced) · PowerPoint · Power BI

Quantitative: econometrics · modelling & forecasting · impact assessment · cost-benefit analysis

Specialisation: sectoral regulation · competition · public service obligations · tariff & cost allocation · compensation / subsidies

Languages: French C1 · Spanish (native) · English C1 · Catalan (basic)

EXPERIENCE

Senior Specialist — Regulation & Competition · Osinergmin (energy regulator, Peru) 2019–2024

- Defended interests in 50+ competition disputes: quantified economic analysis, market power and valuation.
- Quantitative analysis and economic modelling for formal proceedings before regulatory bodies.
- Pedagogical communication of results to senior management and technical committees.

Specialist — Policy & Economic Analysis · Osinergmin · Directorate of Policy & Economic Analysis 2014–2019

- Economic costing and impact evaluation of public service obligations and their compensation (FOSE and MCTER subsidy schemes).
- Tariff design: rate structure, cost allocation and revenue requirements, linked to regulatory accounting.
- Economic models for forecasting and impact studies; international benchmarking.

Macroeconomic Policy Specialist · Ministry of Economy & Finance (MEF), Peru 2012–2013

- Macroeconomic modelling and forecasting (VAR models) in support of public decision-making.

EDUCATION

MSc Economics, Regulation & Competition — Universitat de Barcelona (2025–2026)

MSc Data Science — La Salle – URL, Barcelona (2025–2026)

MSc Behavioral Economics — Paris School of Economics (2018)

MSc Industrial Economics — Toulouse School of Economics (2005–2007)

BSc Economics — PUCP, Peru (2000–2004)

WORKING PAPERS

ELECTRICITY & ENERGY MARKETS

1. The Price of Hedging: Forward Risk Premia in Iberian Power through the Energy Crisis, and What They Cost the Non-Integrated Retailer (2019–2024).
github.com/cmldata/mibel-derivatives/blob/main/paper_derivatives/derivatives_paper.pdf
2. Screening the Firm Resource: De-rating Calibration, Adverse Selection and Lock-in in Spain's 2026/2027 Capacity Mechanism (SA.112483) — An Admissible-Space Audit with Hourly ELCC Data (2019–2024).
github.com/cmldata/capacity/blob/main/paper/cm_spain_paper.pdf

3. Exit and Absorption in Spanish Electricity Retail through the Energy Crisis (2019–2024).
github.com/cmvddata/model_electricity/blob/main/retail_paper.pdf
4. No Firm Capacity, Three Markets: Solar and the Capacity Mechanism in Southern Europe.
github.com/cmvddata/southern-europe-renewables/blob/main/paper/three_markets.pdf
5. The Wrong Index: Concentration in the Binding Resource (Firm Capacity) in the Spanish Wholesale Market.
github.com/cmvddata/papers/blob/main/firm_concentration_paper.pdf

TRANSPORT

1. The Premium-Calibration Sensitivity of an Intermodal Pricing Ceiling: A Theory-Grounded Yield-Management Engine for the BCN–MAD Corridor.
github.com/cmvddata/papers/blob/main/yield_bcn_mad_paper.pdf
2. Where Did the Booking Fences Go? Advance-Purchase Pricing in a Post-Entry Air Duopoly on the Madrid–Barcelona Corridor.
github.com/cmvddata/papers/blob/main/pricing_paper.pdf
3. A Corridor Loses Both of Its Low-Cost Operators: Prices, Passengers, and a Registered Prediction.
github.com/cmvddata/papers/blob/main/avlo_preres.pdf
4. Pricing Response After the Low-Cost Brand Leaves: Independent High-Speed Fares on Madrid–Barcelona When AVLO Withdrew.
github.com/cmvddata/papers/blob/main/railprice_paper.pdf
5. Subsidising the Metro per Passenger or per Kilometre? A Calibrated Translog Cost Function for the Barcelona Metro (FMB-TMB), 2023–2025.
github.com/cmvddata/TMB-costos/blob/main/paper/paper_translog_fmb.pdf
6. The Generalised Price of the Madrid–Barcelona Corridor: What Realised Market Data Reveal About the Air–Rail Trade-off.
github.com/cmvddata/papers/blob/main/genprice_paper.pdf

DATA SCIENCE / MACHINE LEARNING

1. Detecting Structural Manipulation in Electricity Markets: A Supervised Approach Using Partial Regulatory Ground Truth.
github.com/cmvddata/neuro/blob/main/paper/main.pdf
2. Replicating LEAR on MIBEL through the Solar Transition: Why Technical Indicators Do Not Transfer, and Why Percentage Errors Mislead (2022–2024).
github.com/cmvddata/mibel-forecasting/blob/main/paper_forecasting/forecasting_paper.pdf
3. When Psychopathic Traits Are Not Traits: Aggregation Bias and Big-Five Redundancy in the Prediction of Professional Success.
github.com/cmvddata/psychopathy-success/blob/main/docs/paper.pdf
4. Cross-Sectional Characterization of High-Cost Healthcare Users in the US: How the Morbidity-Cost Gradient Varies by Insurance Type in MEPS 2022.
github.com/cmvddata/MEPS_analysis/blob/main/paper/meps_paper.pdf

PUBLIC ECONOMICS · HEALTH · PSYCHOMETRICS

1. Energy Poverty and the Targeting of Electricity Subsidies: Evidence from the FOSE in Peru. (*Master's thesis*)
github.com/cmvddata/fose_analysis/blob/main/docs/paper_fose.pdf
2. Targeting Energy Poverty: Structural Predictors versus Income Thresholds in Spain's Social Electricity Bonus.
github.com/cmvddata/bono-social-microsim/blob/main/paper/paper.pdf
3. Ordinal Distortion of the Spanish Regional Financing System: Quantitative Evidence for the Common-Regime Autonomous Communities, 2009–2022.
github.com/cmvddata/sfa-distorsion-ordinal/blob/main/paper/paper.pdf

CULTURE & SPORTS

1. The Red and the Black: Colombia, Stendhal, and the Price of Arriving Too Soon.
github.com/cmvddata/culture-sports/blob/main/red_and_the_black.pdf
2. Who Drew the Line? Barcelona '92, the Dream Team, and the Industrial Organization of Painted Lines.
github.com/cmvddata/culture-sports/blob/main/who_drew_the_line.pdf
3. You Can't ABX a Memory: New Order's "Bizarre Love Triangle," Glastonbury 1987 vs 2016.
github.com/cmvddata/culture-sports/blob/main/you_cant_abx_a_memory.pdf

4. Was Cumbia Villera Invented in a Yamaha Office in Hamamatsu, Japan? An Aesthetics of Limitation: Technological Democratization, the "Fertile Error," and Sonic Identity in Cumbia Villera.
github.com/cmvddata/culture-sports/blob/main/cumbia_villera_hamamatsu.pdf
5. What Does a Country Buy When It Buys Back a Ball? Relics, Liturgy, and Sixty Years of Hurt: the 1966 Ball and the Symbolic Economy of English Football.
github.com/cmvddata/culture-sports/blob/main/buying_back_a_ball.pdf
6. Mexico '86, Marx and the Hand of God: How a Symbol Makes a Price and an Enemy in the Same Gesture.
github.com/cmvddata/culture-sports/blob/main/mexico86_hand_of_god.pdf
7. At the End, Totò Always Scores: Chance, Merit, and the Italian Summer That Chose Schillaci.
github.com/cmvddata/culture-sports/blob/main/at_the_end_toto_always_scores.pdf
8. Answered to Its Face: The Haka and the Open Game.
github.com/cmvddata/culture-sports/blob/main/haka_open_game.pdf
9. W/kg on a Climb is a Distribution, not a Verdict: Estimating Cycling Climbing Power with Honest Uncertainty.
github.com/cmvddata/cycling/blob/main/wkg/paper.pdf
10. Relaxing the Offside Law in Association Football: A Game-Theoretic, Agent-Based Simulation of Attack–Defense Equilibria.
github.com/cmvddata/offside/blob/main/offside_paper.pdf

PUBLISHED & INSTITUTIONAL

Co-editor of an Osinergmin volume on the Peruvian electricity industry (2016) and co-author of two working papers on energy-sector regulation (Working Papers 32 and 40, 2014–2016). References available on request.